

Cambridge-Aachen Algorithm

List of vectors

Compute pairwise distance

$$d_{ij} = \frac{D(\vec{v}_i, \vec{v}_j)^2}{R^2}, \quad d_{iB} = 1$$

Find the global minimum

$$d_{min} = \min\{\text{all } d_{ij}, d_{iB}\}$$

d_{min} is?
which type?

d_{ij} (vector pair)

Merge vectors

$$\vec{v}_{new} = \frac{\vec{v}_i + \vec{v}_j}{2}$$

New proto-vector replaces pair

d_{iB}

Promote $v_i \rightarrow$ cluster

Remove v_i from the list and add to
final cluster collection

No

Vectors remaining?

Is the proto-vector list empty?

Yes $\rightarrow$ done

Final cluster
collection